

11.28

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

$$\text{Demand: } Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di} \quad (11.11)$$

$$\text{Supply: } Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si} \quad (11.12)$$

- a. Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?

$$\text{Demand: } P_i = -\frac{\alpha_1}{\alpha_2} + \frac{1}{\alpha_2} Q_i - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i - \frac{1}{\alpha_2} e_{di} \quad (1)$$

$$\text{Supply: } P_i = -\frac{\beta_1}{\beta_2} + \frac{1}{\beta_2} Q_i - \frac{\beta_3}{\beta_2} PF_i - \frac{1}{\beta_2} e_{si} \quad (2)$$

anticipated signs of the parameters:

Demand:

$$\alpha_2 < 0, \alpha_3 > 0, \alpha_4 > 0$$

Supply:

$$\beta_2 > 0, \beta_3 < 0$$

- b. Using the data in the file truffles, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?

```
load("/Users/liaochenghao/Desktop/Financial Econometrics/Homework/HW7/Data/truffles.rdata")
df <- truffles

# Demand equation by 2SLS
demand_2sls <- ivreg(
  p ~ q + ps + di |
  ps + di + pf,
  data = df
)
```

```

# Supply equation by 2SLS
supply_2sls <- ivreg(
  p ~ q + pf |
  pf + ps + di,
  data = df
)

# Signs of Demand equation
# Demand one-sided tests
demand_coef <- summary(demand_2sls)$coefficients
df_d <- demand_2sls$df.residual
# H1: Q < 0
p_Q_demand <- pt(demand_coef["q", "t value"], df = df_d)
# H1: PS > 0
p_PS_demand <- 1 - pt(demand_coef["ps", "t value"], df = df_d)
# H1: DI > 0
p_DI_demand <- 1 - pt(demand_coef["di", "t value"], df = df_d)

p_Q_demand

```

```
## [1] 0.01576752
```

```
p_PS_demand
```

```
## [1] 0.002291114
```

```
p_DI_demand
```

```
## [1] 2.337618e-05
```

```

# Signs of Supply equation
# Supply one-sided tests
supply_coef <- summary(supply_2sls)$coefficients
df_s <- supply_2sls$df.residual
# H1: Q > 0
p_Q_supply <- 1 - pt(supply_coef["q", "t value"], df = df_s)
# H1: PF > 0
p_PF_supply <- 1 - pt(supply_coef["pf", "t value"], df = df_s)

p_Q_supply

```

```
## [1] 6.605827e-14
```

```
p_PF_supply
```

```
## [1] 0
```

p-value of testing significance:
Demand:

$$H_a: \frac{1}{\alpha_2} < 0, -\frac{\alpha_3}{\alpha_2} > 0, -\frac{\alpha_4}{\alpha_2} > 0$$

Supply:

$$H_a: \frac{1}{\beta_2} > 0, -\frac{\beta_3}{\beta_2} > 0$$

Using two-stage least squares, the estimated demand equation has the expected signs. The coefficient on Q is negative, while the coefficients on PS and DI are positive. The one-sided p-values are 0.0158 for Q, 0.0023 for PS, and 0.000023 for DI. Therefore, all three coefficients are statistically significant at the 5% level, and PS and DI are significant at the 1% level.

For the supply equation, the estimated coefficients also have the expected signs. The coefficient on Q is positive, and the coefficient on PF is positive. The one-sided p-values are approximately 6.61e-14 for Q and essentially 0 for PF. Thus, both coefficients are highly statistically significant.

Overall, the 2SLS results support the expected signs of both the demand and supply equations, and the estimated coefficients are significantly different from zero.

c. Estimate the price elasticity of demand “at the means” using the results from (b).

```
# coefficient of Q in the 2SLS demand equation
delta2_hat <- coef(demand_2sls)["q"]

# sample means of P and Q
P_bar <- mean(df$p, na.rm = TRUE)
Q_bar <- mean(df$q, na.rm = TRUE)

# price elasticity of demand at the means
elasticity_demand <- (1 / delta2_hat) * (P_bar / Q_bar)
elasticity_demand
```

```
##          q
## -1.272464
```

d. Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI, PS, and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.

```
# Part (d): Sketch supply and demand curves

# Fixed exogenous variables
DI_star <- 3.5
PF_star <- 23
PS_star <- 22

# Coefficients from 2SLS
d_coef <- coef(demand_2sls)
s_coef <- coef(supply_2sls)

# Demand curve: P_D = delta1 + delta2 Q + delta3 PS* + delta4 DI*
d_intercept <- d_coef["(Intercept)"] + d_coef["ps"] * PS_star + d_coef["di"] * DI_star
```

```

d_slope <- d_coef["q"]

# Supply curve:  $P_S = \theta_1 + \theta_2 Q + \theta_3 PF$ 
s_intercept <- s_coef["(Intercept)"] + s_coef["pf"] * PF_star
s_slope <- s_coef["q"]

# Q range
Q_grid <- seq(
  min(truffles$q, na.rm = TRUE),
  max(truffles$q, na.rm = TRUE),
  length.out = 200
)

# Predicted prices
P_demand <- d_intercept + d_slope * Q_grid
P_supply <- s_intercept + s_slope * Q_grid

# Equilibrium point
Q_eq <- (s_intercept - d_intercept) / (d_slope - s_slope)
P_eq <- d_intercept + d_slope * Q_eq

# Plot
plot(
  Q_grid, P_demand,
  type = "l",
  lwd = 2,
  xlab = "Quantity (Q)",
  ylab = "Price (P)",
  main = "Estimated Demand and Supply Curves"
)

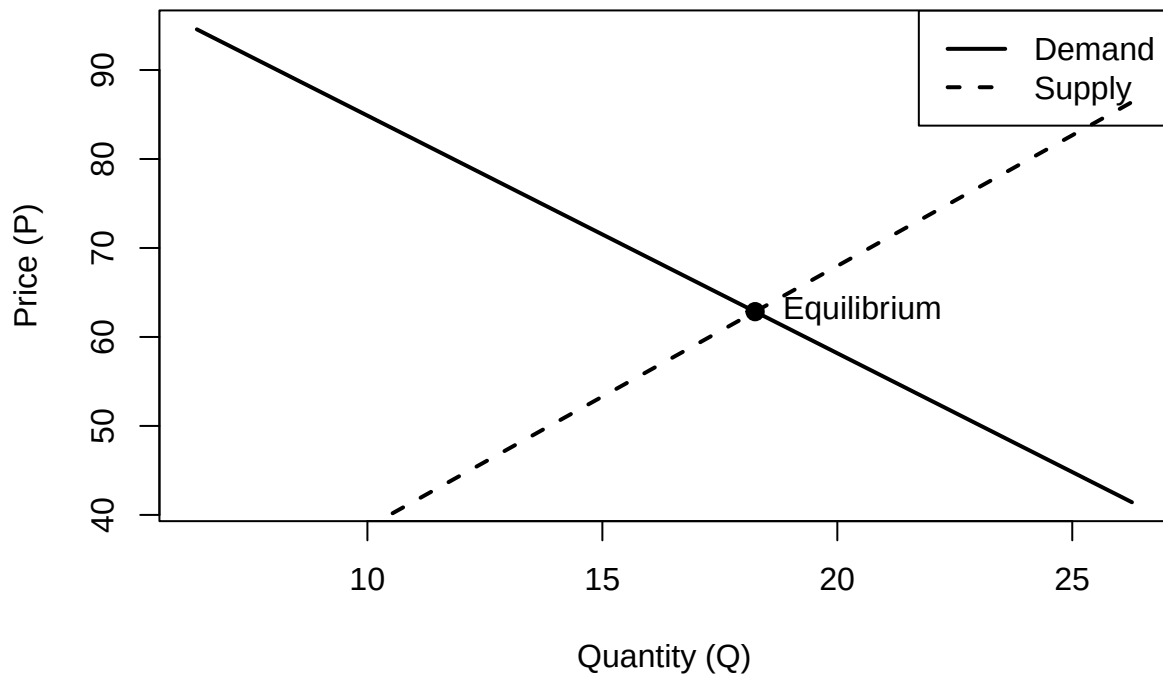
lines(Q_grid, P_supply, lwd = 2, lty = 2)

points(Q_eq, P_eq, pch = 19, cex = 1.2)
text(Q_eq, P_eq, labels = " Equilibrium", pos = 4)

legend(
  "topright",
  legend = c("Demand", "Supply"),
  lwd = 2,
  lty = c(1, 2)
)

```

Estimated Demand and Supply Curves



e. What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?

Q_{eq}

```
## (Intercept)
##      18.25021
```

P_{eq}

```
## (Intercept)
##      62.84257
```

Reduced-form coefficients from Table 11.2

```
PS_star <- 22
DI_star <- 3.5
PF_star <- 23
```

Reduced form for Q

```
Q_eq_reduced <- 7.8951 + 0.6564 * PS_star + 2.1672 * DI_star - 0.5070 * PF_star
```

Reduced form for P

```
P_eq_reduced <- -32.5124 + 1.7081 * PS_star + 7.6025 * DI_star + 1.3539 * PF_star
```

```
Q_eq_reduced
```

```
## [1] 18.2601
```

```
P_eq_reduced
```

```
## [1] 62.81425
```

```
comparison <- data.frame(  
  Method = c("Structural equations from part (d)", "Reduced-form equations from Table 11.2"),  
  Q = c(Q_eq, Q_eq_reduced),  
  P = c(P_eq, P_eq_reduced)  
)
```

```
comparison
```

```
##                               Method      Q      P  
## (Intercept)      Structural equations from part (d) 18.25021 62.84257  
##                Reduced-form equations from Table 11.2 18.26010 62.81425
```

Using the estimated structural equations from part (d), the equilibrium values are $Q = 18.25021$ and $P = 62.84257$.

Using the reduced-form equations from Table 11.2 and substituting $PS = 22$, $DI = 3.5$, and $PF = 23$, the predicted values are $Q = 18.26010$ and $P = 62.81425$.

The two sets of values are very close. The difference in Q is only about 0.00989, and the difference in P is only about 0.02832. Therefore, the equilibrium values obtained from the structural equations and the reduced-form equations agree very well. The small difference is probably due to rounding of the reported coefficients.

-
- f. Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

```
# Demand equation by OLS  
demand_ols <- lm(p ~ q + ps + di, data = df)  
  
# Supply equation by OLS  
supply_ols <- lm(p ~ q + pf, data = df)  
  
# Results  
coef(demand_ols)
```

```
## (Intercept)      q      ps      di  
## -13.6194989  0.1512043  1.3607045 12.3582353
```

```
coef(supply_ols)
```

```
## (Intercept)          q          pf  
## -52.876298    2.661281    2.921660
```

```
# Signs of Demand equation
```

```
# Demand one-sided tests
```

```
demand_coef <- summary(demand_ols)$coefficients
```

```
df_d_f <- demand_ols$df.residual
```

```
# H1: Q < 0
```

```
p_Q_demand <- pt(demand_coef["q", "t value"], df = df_d_f)
```

```
# H1: PS > 0
```

```
p_PS_demand <- 1 - pt(demand_coef["ps", "t value"], df = df_d_f)
```

```
# H1: DI > 0
```

```
p_DI_demand <- 1 - pt(demand_coef["di", "t value"], df = df_d_f)
```

```
p_Q_demand
```

```
## [1] 0.6179094
```

```
p_PS_demand
```

```
## [1] 0.01516251
```

```
p_DI_demand
```

```
## [1] 1.740691e-07
```

```
# Signs of Supply equation
```

```
# Supply one-sided tests
```

```
supply_coef <- summary(supply_ols)$coefficients
```

```
df_s_f <- supply_ols$df.residual
```

```
# H1: Q > 0
```

```
p_Q_supply <- 1 - pt(supply_coef["q", "t value"], df = df_s_f)
```

```
# H1: PF > 0
```

```
p_PF_supply <- 1 - pt(supply_coef["pf", "t value"], df = df_s_f)
```

```
p_Q_supply
```

```
## [1] 2.664535e-15
```

```
p_PF_supply
```

```
## [1] 0
```

Using OLS, the estimated demand equation does not fully satisfy the expected signs. The coefficients on PS and DI are positive and statistically significant, with one-sided p-values of 0.0152 and 1.74e-07, respectively. However, the coefficient on Q is not significantly negative. Its one-sided p-value is 0.6179, so the OLS demand equation does not provide evidence of a downward-sloping demand curve.

For the supply equation, the OLS results have the expected signs. The coefficient on Q is positive and highly significant, with a p-value of 2.66e-15. The coefficient on PF is also positive and highly significant, with a p-value essentially equal to zero.

Compared with the 2SLS results in part (b), the OLS results are less satisfactory. In part (b), the 2SLS demand equation had the expected signs and all coefficients were statistically significant. In contrast, the OLS demand equation fails to produce a significantly negative coefficient on Q .

The reason is that quantity is endogenous in the structural supply and demand equations. Since price and quantity are jointly determined in equilibrium, OLS suffers from simultaneity bias. Therefore, the 2SLS estimates are preferred to the OLS estimates.